# _DERIVATIONS: AN AUTONOMOUS SYSTEM FOR RESPONSIVE LIVE MUSIC IMPROVISATION

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**ABSTRACT**

Much recent research in the area of interactive computer music is concerned with the creation of computer systems that respond dynamically to musical input provided by improvising instrumentalists. These systems have the capacity to participate both responsively and autonomously in a live performance context, acting as virtual improvisers imbued with an awareness of the current and past musical context provided by one or more human improvisers. Various systems of this type strive for novelty and spontaneity in their responses, but also aim to coherently integrate computer-produced sounds with those of the musician. This research examines practical and aesthetic motivations behind systems that derive their sonic and algorithmic responses directly from an instrumentalist's live performance. The current iteration of the author's *_derivations* system is used as a case study to examine the design approach and contextualize this work in relation to earlier works by the author and relevant recent work of others in this field, particularly works that focus on *timbre* and sonority derived from the live instrument, rather than pattern or rule-based melodic invention.

Interactive systems have been described in terms of their ability to meet the musical aesthetics of their system designers (Drummond 2009). The design of sonic responses within performative systems is also dependant upon the musical and sonic context envisaged for the live interaction. In the *Omax-Ofon* system (Assayag, Bloch and Chemillier 2006), interaction is achieved by presenting the instrumentalist with a clone of himself, with the system referencing a continuously updated repository time-stamped and descriptor-analysed recordings of the performance using pattern matching techniques. In Eldridge's *Fond Punctions* (Eldridge 2005, 2008), performer triggered live sampling and generatively (by Neural Oscillator Networks) controlled granular synthesis techniques create 'timbral reflections' of the sampled improvisations. In Hsu's work with improvisatory systems, the focus is on the analysis and storage of timbral characteristics of the sound produced by a human improviser, and how this data can control and influence the behaviour modes of the software agents interacting with the performer (Hsu 2005, 2006, 2008, 2009).

In designing system response in the current iteration of the author's work, *_derivations*, a central aesthetic concern was to create a unified timbral palette that blends computationally generated sound with the live instrumental sound, i.e. the human and machine must cohabit in the same, unified sound world. The system response is designed to augment and transform the sonic environment in which it is operating, organically feeding off the environment provided by the performer's musical gestures. To achieve this, the system has access to all that has previously been played by the instrumentalist to derive its sonic responses. *_derivations* employs a combination of continuous live sampling of the instrumental performance, as well the analysis and storage of timbral information captured from the live signal. In the current design, three synthesis modules – a granulator, a bank of four phase vocoders and an additive re-synthesis module - reference phrase-analysed segments of recorded audio and spectral data to generate their sonic material. Current transformation techniques are based on quasi-aleatoric processes devised by the author. It is suggested that the unique combination of analysis, storage and transformation techniques present in the current design provides a rich and complex environment for human-machine interaction, and fertile ground for ongoing research.

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